



Enhancing keratoconus detection with transformer technology and multi-source integration

Osama Ismael¹

Accepted: 6 November 2024 / Published online: 21 November 2024
© The Author(s) 2024

Abstract

Keratoconus is a progressive eye disease characterized by the thinning and conical distortion of the cornea, leading to visual impairment. Early and accurate detection is essential for effective management and treatment. Traditional diagnostic methods, relying primarily on corneal topography, often fail to detect early-stage keratoconus due to their subjective nature and limited scope. In this research, we present a novel multi-source detection approach utilizing transformer technology to predict keratoconus progression more accurately. By integrating and analyzing diverse data sources, including corneal topography, aberrometry, pachymetry, and biomechanical properties, our method captures subtle changes indicative of disease progression. Transformer networks, known for their capability to model complex dependencies in data, are employed to handle the multimodal datasets effectively. Experimental results demonstrate that our approach significantly outperforms existing methods, such as SVM-based, Random Forests-based, and CNN-based models, in terms of accuracy, precision, recall, and F-score. Moreover, the proposed system exhibits lower execution times, highlighting its efficiency in clinical settings. This innovative methodology holds the potential to revolutionize keratoconus management by enabling earlier and more precise interventions, ultimately enhancing patient outcomes and contributing significantly to both the medical and machine learning communities.

Keywords Keratoconus progression · Transformers technology · Multi-source data integration · Deep learning · Corneal topography

Abbreviations

AI	Artificial intelligence
AS-OCT	Anterior segment-optical coherence tomography
CNN	Convolutional neural network
FFNN	Feed-forward neural network
NLP	Natural language processing

✉ Osama Ismael
osama@fci-cu.edu.eg

¹ Faculty of Computers and Artificial Intelligence, Cairo University, Cairo, Egypt

OCT	Optical coherence tomography
PCA	Principal component analysis
SVM	Support vector machine
ViT	Vision transformer

1 Introduction

Keratoconus is a progressive, non-inflammatory corneal disease characterized by corneal thinning, conical distortion, and protrusion of the cornea, leading to significant visual impairment and, in severe cases, blindness (Usgaonkar et al. 2023). The early and accurate detection of keratoconus progression is critical to preventing vision loss and treating the disease effectively. Traditional diagnostic methods, primarily relying on corneal topography, often focus on morphological changes in the cornea, which can be subjective and time-consuming. However, these methods often struggle with detecting early-stage keratoconus due to the subtle nature of the early morphological changes, leading to delayed intervention and suboptimal treatment outcomes.

On the other hand, in recent years, machine learning techniques have emerged as promising tools for improving the accuracy and efficiency of Keratoconus diagnosis and progression prediction. These techniques have employed various algorithms, such as Support Vector Machines (SVM), Random Forests, and Convolutional Neural Networks (CNNs). SVM have been widely used, utilizing handcrafted features from corneal topography data to distinguish between normal and keratoconic eyes. Studies like those by Lavric et al. (2020), and Shanthi et al. (2020) employed SVM to classify keratoconus based on corneal shape, thickness, and elevation profiles. While these methods have shown moderate success, they rely heavily on manual feature engineering and often lack the capability to generalize across varying datasets and patient populations. Other approaches, such as Random Forest models, have also been explored. Cohen et al. (2022) developed a Random Forest-based system to analyze corneal tomography images, demonstrating promising accuracy. However, similar to SVM, Random Forest models are limited in their ability to integrate multimodal data sources, which are critical in capturing the full spectrum of keratoconus progression.

Moreover, deep learning techniques, a subfield of machine learning, have the ability to automatically learn features from data. This eliminates the need for manual feature engineering and can potentially lead to improved performance. Deep learning algorithms, particularly CNNs, as applied by Wan et al. (2023), Kallel et al. (2024), and others, have shown significant promise for Keratoconus progression detection using corneal topography images. CNNs automatically learn features from image data, eliminating the need for manual feature extraction. Despite these advances, CNNs primarily handle image-based data and struggle to capture the complex, multimodal nature of keratoconus diagnosis, which involves integrating data from multiple diagnostic tools, such as aberrometry, pachymetry, and biomechanical measurements.

Given the limitations of these existing methods in handling diverse data sources, and detecting and monitoring early-stage keratoconus, there is a pressing need for a more comprehensive and objective approach (Niazi et al. 2023). This research aims to address this gap by developing a novel transformer-based model for keratoconus detection that integrates

multimodal data from corneal topography, aberrometry, pachymetry, and biomechanical properties, hence extracting meaningful patterns that are indicative of disease progression.

Transformer networks, a relatively new deep learning architecture, have achieved state-of-the-art results in various Natural Language Processing (NLP) tasks (Thomas et al. 2020). Transformers, with their self-attention mechanisms, excel at capturing complex and long-range dependencies within data, making them well-suited for tasks involving sequential or multimodal data. Moreover, there are numerous studies on multi-source data fusion and transformer models for various prediction tasks. For instance, transformer-based methods have been effectively utilized for tasks like wind power forecasting, where they integrate various data sources to enhance prediction accuracy (Liu et al. 2023). Additionally, transformer models have been used for fault diagnosis and condition evaluation in industrial applications, highlighting their robustness in handling complex, multi-source datasets (Zhang et al. 2024). By leveraging transformers, our model can detect subtle signs of keratoconus, including early-stage or subclinical cases, with higher accuracy and efficiency than existing methods. While there is a growing body of research on deep learning for keratoconus progression detection, there are no studies investigating the use of transformers for this specific task. This highlights an opportunity for further research to explore the potential of transformers in improving the accuracy and efficiency of multi-source keratoconus progression detection. The innovations of this research manifest can be summarized in the following points:

1. **Application to a novel domain:** While transformers have been predominantly used in areas such as natural language processing and image analysis, their application in the diagnosis and progression prediction of keratoconus is relatively unexplored. To the best of our knowledge, this is one of the first studies to apply transformer networks for the integration of corneal topography, aberrometry, pachymetry, and biomechanical data, all crucial in detecting subtle signs of keratoconus. This innovative use case extends the applicability of transformer models beyond traditional domains.
2. **Multimodal data fusion:** One of the core innovations lies in how the transformer model is utilized to integrate diverse types of data that have traditionally been analyzed separately in keratoconus studies. The model efficiently handles both spatial and temporal dependencies across corneal imaging (topography, pachymetry) and biomechanical properties, offering a more holistic approach to keratoconus detection. This is a novel approach in ophthalmology, addressing a critical gap in current diagnostic models that rely heavily on single data sources, often leading to inconsistent results.
3. **Early detection and subclinical keratoconus classification:** Existing diagnostic methods struggle with detecting early-stage or subclinical keratoconus due to the subtle nature of the early morphological changes. Our work leverages the high sensitivity of transformers to model complex dependencies across multimodal datasets, allowing for more accurate early detection and differentiation between normal, suspected, and definite keratoconus cases. This contributes to the field by providing a framework capable of identifying keratoconus earlier than conventional methods.
4. **Custom architecture and model optimization:** While the transformer is a mature framework, our study introduces optimizations tailored for the keratoconus detection task. This includes specialized preprocessing techniques for handling noisy, high-dimensional ophthalmic data, as well as feature fusion strategies that enhance the model's

ability to handle heterogeneous data types. These architectural adjustments are key to achieving the high levels of accuracy and efficiency demonstrated in the results, particularly when compared to traditional machine learning models such as SVMs and CNNs.

5. **Clinical relevance and scalability:** The proposed system is not only designed for accuracy but also for clinical efficiency. By demonstrating lower execution times and superior scalability, the transformer-based model is poised for integration into real-world clinical environments where rapid and reliable keratoconus detection is needed. This aspect of innovation makes the proposed approach more feasible for real-time applications in healthcare settings, addressing both technological and practical challenges.

As a conclusion, the innovation in this study lies in the novel application of this architecture to keratoconus detection, the integration of multimodal datasets, and the development of a model optimized for both early detection and clinical deployment. Additionally, it could serve as a foundation for future research in other corneal diseases and ocular conditions (Wang et al. 2024), and it can contribute to the development of novel transformer-based approaches for medical image analysis and multimodal data integration.

2 Literature review

Several studies have explored the use of machine learning algorithms for Keratoconus progression detection. These studies have employed various algorithms, including Support Vector Machines (SVM) (Alyaa et al. 2018; Lavric et al. 2020; Shanthi et al. 2020), Random Forests (Cohen et al. 2022), and other machine learning techniques (Yousefi et al. 2018; Kamiya et al. 2019, 2021). These algorithms frequently experience poor detection accuracy because of the lack of different diagnostic sources, such as angiography, ultrasonography, pachymetry, optical coherence tomography, computed tomography, and magnetic resonance imaging. In addition, deep learning has revolutionized various domains of medical imaging, including ophthalmology. Several studies have demonstrated the efficacy of deep learning in detecting and classifying ocular diseases, such as diabetic retinopathy, glaucoma, and age-related macular degeneration. In recent years, the detection and classification of keratoconus progression disease using deep learning techniques, particularly CNNs, has gained significant attention due to its potential for improving early diagnosis and personalized treatment plans (Lavric and Valentin 2019; Kuo et al. 2020; Feng et al. 2021; Chen et al. 2021; Wan et al. 2023; Kallel et al. 2024). However, CNNs are primarily designed for image data and may not be able to fully capture the complex relationships between different data sources that might be relevant for Keratoconus progression detection. Hence, this section reviews the key studies and methodologies that have contributed to this field.

Alyaa et al. (2018) presented a method that relies on information retrieved from topographic maps to detect Keratoconus that affects the cornea in a diseased eye using image processing techniques. They used twelve features retrieved from topographic maps generated by the Pentacam instrument and applied them to the SVM to determine if the cornea was healthy or damaged. Lavric et al. (2020) applied and tested a set of machine learning algorithms to detect keratoconus based on shape, thickness, and elevation profiles of the cornea. They used real-world medical data, including corneal topography, elevation, and

pachymetry parameters, which were collected from OCT-based topography instruments from several corneal clinics in Japan. They used a subset of eight corneal parameters with the highest discriminating power. Based on the experiments, the SVM algorithm achieved the highest accuracy level. Likewise, (Shanthi et al. 2020) proposed a computer-aided diagnosis model based on SVM learning to detect the early stage of keratoconus using the available topographic, pachymetric, and aberrometry parameters of patients with keratoconus, subclinical keratoconus, and normal corneas.

Cohen et al. (2022) developed an autonomous machine-learning model for distinguishing between normal corneas and suspected abnormal or keratoconic corneas. In their experiments, they employed 8526 corneal tomography pictures from 4904 eyes taken with a combined Scheimpflug/Placido tomographer. A cornea specialist analyzed each image's capture quality and classified it as normal, suspicious irregular, or keratoconic. Two algorithms were created utilizing the random forest algorithm; the first used 94 instrument-derived output parameters, whereas the second combined the device's keratoconus prediction indices with the 94 instrument-derived output parameters. Both models were evaluated against the tomographer's keratoconus detection methods.

In order to detect and track keratoconus progresses, (Yousefi et al. 2018) developed an unsupervised machine learning algorithm and applied it to corneal parameters (topographic, tomographic, and thickness profiles of the cornea). They utilized the corneal swept-source Optical Coherence Tomography (OCT) image dataset from the SS-1000 CASIA OCT Imaging Systems. 420 corneal topography, elevation, and pachymetry parameters were selected. There were three main steps in the algorithm. (1) The dimensionality of the input data was linearly reduced from 420 to eight significant principal components using the Principal Component Analysis (PCA) technique. (2) The selected principal components were further reduced nonlinearly to two eigen-parameters using manifold learning. (3) In order to identify eyes with keratoconus, density-based clustering was finally applied to the eigen-parameters. Additionally, the ectasia status index was used to evaluate the accuracy of the detected clusters objectively, and the visualization of clusters in two-dimensional space was utilized to check the learning quality subjectively. Likewise, Kamiya et al. (2019, 2021) suggested a model for predicting keratoconus progression based on deep learning of six colour-coded maps obtained using swept-source Anterior Segment-Optical Coherence Tomography (AS-OCT) to evaluate the accuracy of keratoconus detection. The anterior elevation, anterior curvature, posterior elevation, posterior curvature, total refractive power, and pachymetry map were the six color-coded maps. Additionally, 239 age-matched healthy eyes and 304 keratoconic eyes - grades 1 (108 eyes), 2 (75 eyes), 3 (42 eyes), and 4 (79 eyes) - according to the Amsler-Krumeich classification were employed.

Lavric and Valentin (2019) used the convolutional neural network to analyze the corneal topography of the eye to determine whether an eye is affected or not by keratoconus pathology. Also, Kuo et al. (2020) developed and compared deep learning algorithms to detect keratoconus on the basis of corneal topography and validate with visualization methods. Their research is based on a collection of corneal topographies images, which include clinically manifested keratoconus and regular astigmatism. The images were divided into training and test datasets. They adopted three CNN models for learning. In addition, they displayed the pixel-wise discriminative features and class-discriminative heat map of diopter images for visualization. Based on their experiments, the pixel-wise discriminative features and the heat map of the prediction layer in the VGG16 model both revealed it focused on the

largest gradient difference of topographic maps; this corresponded with ophthalmologists' diagnostic clues.

Feng et al. (2021) proposed an end-to-end deep learning approach called KerNet, based on CNN, to detect keratoconus and subclinical keratoconus. In which they process the raw data of the Pentacam HR system, this data includes the front and back surface curvature, the front and back surface elevation, and the pachymetry of an eye. The suggested model's architecture consists of five branches that serve as the backbone for a multilevel fusion architecture. The five branches receive five slices separately, and several cascaded residual blocks capture the features of different slices. But its main drawback lies in incurring a lot of memory and time costs compared to other state-of-the-art methodologies. Besides, it still cannot determine the subtypes or the severity of keratoconus diseases. Also, Chen et al. (2021) implemented a set of experiments to evaluate the accuracy of the CNN technique in detecting keratoconus using color-coded corneal maps obtained by a Scheimpflug camera. To train and test their CNN models, they used the images of corneal tomography scans of keratoconic and healthy volunteers' eyes provided by three centers: Royal Liverpool University Hospital (UK), Sedaghat Eye Clinic (Mashhad, Iran), and The New Zealand National Eye Center (New Zealand). To classify the keratoconic scans, they used the Amsler-Krumeich classification technique. Each scan was analysed using four maps: an axial map, anterior and posterior elevation maps, and a pachymetry map.

Wan et al. (2023) developed a model based on deep learning techniques to identify and assess the corneal endothelial cell changes and diagnose keratoconus based on corneal endothelial images acquired from a non-contact specular microscope. They used the specular microscopy to image and measure the morphometric parameters of the corneal endothelial cells. They utilized a Fully Convolutional Network model with a ResNet50 (FCN_ResNet50) to perform the endothelial segmentation. The images were then identified utilising an ensemble machine learning system composed of four pre-trained deep learning networks: DenseNet121, ResNet50, Inception_v3, and MobileNet_v2. Also, they constructed an associated nomogram for diagnosis, monitoring, and management of the progression of keratoconus. Similarly, Kallel et al. (2024) introduced a customized CNN model aiming at the classification of corneal topographic maps and the detection of keratoconus disease. They compared their model with six different deep learning models, two hybrid models named CNN-SVM and CNN-LSTM, and four transfer learning models, known as VGG19, Xception, ResNet 50, and MobileNetV2. They used two data augmentation methods, the SyntEyes models and the generative adversarial networks, to generate synthetic corneal topographic maps and increase the size of the training datasets, respectively.

Based on the aforementioned, this literature review provides a brief overview of machine learning and deep learning approaches for keratoconus progression detection. Despite the advancements, several challenges remain in the detection of keratoconus progression using deep learning. One major challenge is the scarcity of large, annotated datasets, which limits the training and validation of robust models. Additionally, the variability in imaging modalities and clinical protocols across different studies poses a challenge for model generalizability. It highlights the limitations of traditional methods and the potential of transformer technology for multi-source detection. Hence, further research is warranted to investigate the efficacy of transformer-based models in this domain and to develop robust and generalizable models for improved clinical decision-making in keratoconus management.

3 Transformers-based multi-source detection

Motivated by the aforementioned issues, we suggest a novel methodology for detecting keratoconus progression that leverages the advanced capabilities of transformer networks to integrate and analyze multi-source data. The methodology is structured into several key components: data preprocessing, data integration, and classifier construction. Each component is critical to ensuring the accuracy and reliability of the proposed system. The data is prepared for the detection process in the first component, and then in the second component, the extracted features will be combined into a unified representation suitable for transformer input. Finally, in the third component, we construct the detection model with the transformers, as shown in Fig. 1. These three components are thoroughly described in the next three subsections.

3.1 Data preprocessing

Data preprocessing is essential for ensuring the quality and consistency of both the training and testing datasets. This component is broken down into three main steps: feature extraction, noise reduction, and data normalization. Data generally contains irrelevant and redundant features; thus, it is essential to identify and extract relevant features from each data source, especially when dealing with multi-source data for tasks like keratoconus progression detection. As a result, feature extraction is an important component of any disease detection system because it can easily identify a subset of the most relevant features and remove unnecessary or redundant features, thereby improving system performance and decreasing computation time. In addition, using all of the features makes the system more complex while decreasing its accuracy. As a result, our goal is to select the most relevant subset of features for the mission at hand. Feature selection is divided into two types: filter methods and wrapper methods. Wrapper methods train a model with a minimal set of features, whereas filter approaches rank features using statistical techniques. Due to their high computational cost, wrapper methods are not the most effective feature selection strategies to consider when working with large datasets. In addition, utilizing a subset of features from wrapper methods makes the model more susceptible to overfitting than using a subset of features from filter methods (Pudjihartono et al. 2022). Hence, the different advantages and disadvantages of each strategy suggest that there is no single optimum method suitable for every situation; rather, it is determined by the specific dataset and methodology objectives. Consequently, we prefer filter methods over wrapper methods since they are significantly faster and require no model training.

The filter method is a popular approach for feature selection, as it evaluates the relevance of features independently of any machine learning model. For a task like keratoconus progression detection, which involves integrating data from various sources, we recommend using the PCA approach (Jolliffe and Cadima 2016). PCA is a powerful dimensionality reduction technique that can be effectively employed in the task of keratoconus progression detection. Given the multi-source nature of the data involved, datasets can often become high-dimensional and complex. PCA assists in converting this high-dimensional data into a lower-dimensional space while retaining as much of the original variance as possible. By projecting the original features onto a set of orthogonal components, PCA reduces the dimensionality of the dataset, making it computationally more efficient for subsequent anal-

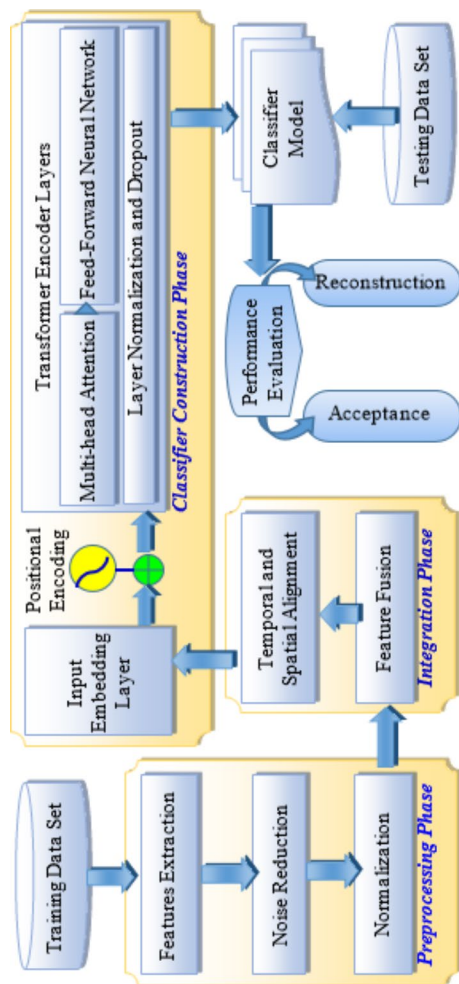


Fig. 1 The transformers-based multi-source keratoconus detection model architecture

ysis. This reduction is particularly beneficial in avoiding the curse of dimensionality, which can degrade model performance due to overfitting when too many features are present. Moreover, PCA can help in uncovering latent patterns and relationships within the data that might not be apparent from the original features. In the context of experiments conducted on the keratoconus detection, the PCA was able to achieve up to 62.5% dimensionality reduction, which not only simplifies the dataset but also enhances the performance and generalizability of the deep learning models by focusing on the most informative aspects of the data, thereby improving the accuracy and reliability of predictions. The process of PCA is summarized in the following steps, illustrating how principal components are derived and their relationship to the original data:

1. Center the data by subtracting the mean of each variable and standardizing them to ensure all variables contribute equally to the analysis.

$$S = \frac{x - \bar{x}}{\sigma}$$

2. Compute the covariance matrix to measure how variables vary together, indicating the strength and direction of their relationships.

$$\text{cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

3. Calculate the eigenvectors (principal components) and eigenvalues of the covariance matrix. Eigenvectors identify the directions of maximum variance, while eigenvalues indicate the magnitude of this variance.
4. Choose a subset of principal components with the highest eigenvalues, as they capture the most significant variance in the data.
5. Recast the original data along the selected principal component axes, reducing dimensionality while retaining most of the information.

On the other hand, noise reduction is a crucial step in preprocessing data for keratoconus progression detection, as it ensures that the data fed into the model is clean and reliable. In this context, noise refers to any irrelevant or extraneous information in the data that can obscure meaningful patterns and degrade model performance. Given the multi-source nature of keratoconus data, two different noise reduction techniques are employed to enhance data quality: The Gaussian filter and PCA techniques. The Gaussian Smoothing Operator is particularly effective in reducing random noise in imaging data (Haddad and Akansu 1991). It averaging the pixel values in a local neighborhood based on the Gaussian distribution, thus smoothing out fluctuations while preserving essential details. As for non-image data, we use the PCA technique, as it is used not only to reduce dimensions but also to eliminate noise by reconstructing the data using only the most significant components, effectively filtering out less informative, noise-prone dimensions. Employing these noise reduction techniques helps in improving the signal-to-noise ratio, thereby enhancing the accuracy and robustness of the model used for detecting keratoconus progression. For example, Fig. 2 shows two corneal topography maps, showing the effect of noise reduction using Gaussian filters. The

left image illustrates the cornea before noise reduction, while the right image shows the cornea after applying Gaussian filtering for noise reduction.

Finally, normalization is an essential preprocessing technique in the keratoconus progression detection task, as it ensures that the data from different sources are on a comparable scale. Given the diverse nature of keratoconus data, which includes metrics such as corneal curvature, aberrometry values, and pachymetry measurements, the ranges and units of these features can vary significantly. Without normalization, models may become biased towards features with larger scales, which could negatively impact their performance. To address this, we utilized the min-max scaling technique (Han et al. 2022). Min-max scaling transforms the data into a fixed range, usually $[0, 1]$, by linearly scaling each feature based on its minimum and maximum values. This approach is particularly useful when the data distribution is not Gaussian and when preserving the original relationships in the data is important. By applying this normalization technique, we ensure that each feature contributes equally to the model's learning process, thereby improving the accuracy and generalizability of the keratoconus progression detection system.

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

Where, $\min_A$ and $\max_A$ are the minimum and maximum values of the feature, A , respectively. Min-max scaling maps a value, v , of A to v' in the range $[\text{new_min}_A \text{ and } \text{new_max}_A]$.

3.2 Data integration

The integration of diverse data sources is a crucial component of the proposed keratoconus detection framework. Since keratoconus is a complex condition that manifests through multiple corneal characteristics, the integration of various modalities—such as corneal topography, aberrometry, pachymetry, and biomechanical properties—is essential for creating a comprehensive and accurate model. While transformers are powerful tools for processing data once it is in the correct format, they do not inherently integrate data from multiple sources. Instead, they are designed to work with structured input data. Therefore, it's important to ensure that the input data is preprocessed, correctly integrated, and transformed into the required format before feeding it into a transformer. This preprocessing is crucial for

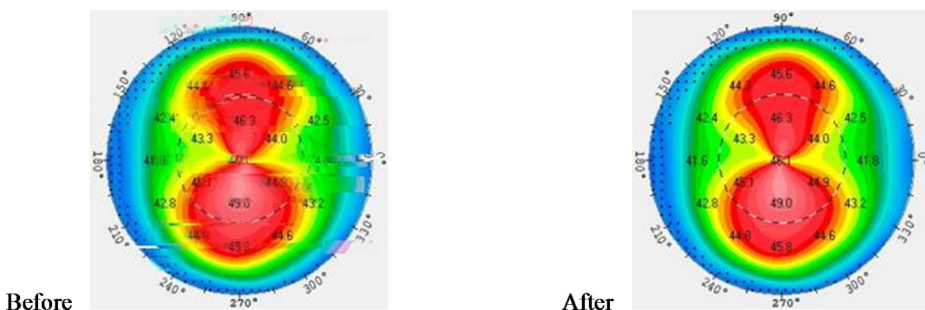


Fig. 2 The corneal topography map before noise reduction in the left image and the corneal topography map after applying Gaussian filtering for noise reduction in the right image

achieving accurate and reliable results in the analysis or deep learning tasks. Hence, in the proposed approach, the data integration process consists of the following key steps:

1. **Feature fusion:** combining features from different sources into a unified representation, which enables the model to capture the intricate relationships and complementary information inherent in the data. Transformers, with their self-attention mechanisms, are particularly well-suited for this task as they can effectively model the dependencies between features across different data types. The process of feature fusion involves aligning and integrating the features from each source, either by concatenating them into a single feature vector or by using the multi-head attention method to learn the optimal feature interactions (Vaswani et al. 2017). Features from different modalities are concatenated into a single, high-dimensional vector. This concatenated vector serves as the input to the transformer model, allowing it to process a comprehensive representation of the corneal state. The multi-head attention mechanism within transformers is then employed to dynamically learn the relationships and interactions between the different features, allowing the model to focus on the most informative aspects of each data source, enhancing the predictive capability of the model.
2. **Temporal and spatial alignment:** temporal alignment involves synchronizing data collected from multiple sources over time to ensure that each time point reflects the same physiological state of the eye so that they correspond to the same temporal instance in a patient's monitoring timeline. Given the longitudinal nature of keratoconus progression, where measurements may be taken at different intervals and frequencies, temporal alignment guarantees temporal consistency in the data inputs to the model. This is particularly important when using transformer models, which rely on attention mechanisms to capture temporal patterns and relationships within the data. To achieve this, two techniques were utilized: dynamic time warping and linear interpolation, to harmonize data from different sources, allowing for a unified temporal framework. Dynamic time warping is particularly effective in aligning time-series data that may have been collected at irregular intervals or contain varying lengths, as it minimizes temporal distortions by mapping data points from different sequences based on their similarity (Meinard 2007). Linear interpolation, on the other hand, was used to fill in missing data points by estimating values between known observations, ensuring a continuous and uniform time-series dataset [Linear interpolation, Encyclopedia of Mathematics]. Spatial alignment is also considered, especially when dealing with images (e.g., topography maps). Here, we ensure that features extracted from different spatial regions of the cornea are accurately aligned and mapped to the same coordinates to maintain consistency across the dataset. By aligning the data temporally and spatially, the model can better understand the progression patterns of keratoconus, enabling it to detect changes more accurately and predict future developments.
3. **Unified representation for classification:** The outcome of this integration process is a unified, multimodal representation of the corneal data. This representation captures the comprehensive corneal profile, reflecting both global and local changes across multiple diagnostic modalities. This integrated data is then passed through the transformer model, which, using its self-attention mechanism, processes the complex relationships within this unified representation to classify the corneal state into categories such as normal, suspected keratoconus, or definite keratoconus.

The process of the data integration is illustrated in Fig. 3. This unified representation allows the transformer to attend to the most relevant aspects of the data, leading to a more holistic understanding of the patient's condition. By fusing features from various sources, the model gains a comprehensive view of keratoconus progression, thereby enhancing its ability to detect subtle changes and predict disease progression more accurately.

By integrating diverse data sources in this manner, the model gains a holistic understanding of the corneal condition, leveraging complementary information from each modality to enhance the accuracy and robustness of keratoconus detection. This integration process is central to the success of the transformer-based approach, as it enables the model to make nuanced distinctions between different stages of the disease.

3.3 Transformers-based classifier construction

The proposed classifier model for detecting keratoconus progression leverages the powerful transformer architecture, which is renowned for its ability to handle complex, multi-modal data. This design is centered around integrating diverse data sources, providing a comprehensive approach to understanding and detecting keratoconus progression. The transformer-based classifier model is specifically tailored to capture intricate patterns and relationships within this heterogeneous data, facilitating accurate and timely detection of disease progression. There are three architecture types for designing a transformer network:

- Vision Transformer (ViT): for handling image-based data (e.g., corneal topography).
- Transformer encoder: for processing sequential data (e.g., time series of measurements).
- Hybrid architecture: combining both ViT and Transformer Encoder for multimodal data integration.

Therefore, based on the complexity of the data and computational resources, the proposed methodology will be based on the hybrid transformer architecture, which will be a suitable architecture to address the research gap under study. Hence, the proposed classifier model will consist of multiple layers that enable the model to effectively process and integrate the multi-source data inputs. The primary components of the proposed transformer architecture model will include the following layers: input embedding layer, positional encoding, and transformer encoder layers.

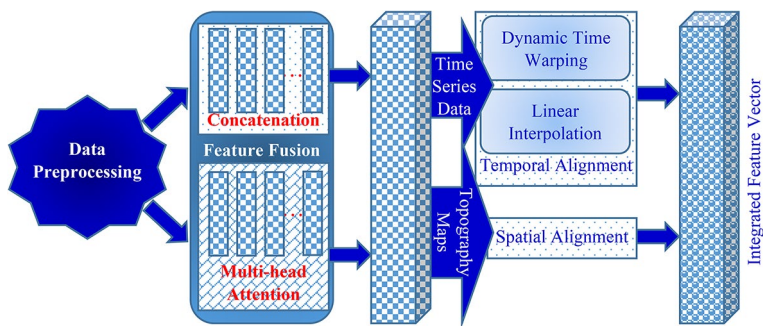


Fig. 3 The data integration process

3.3.1 Input embedding layer

We designed a specialized input embedding layer tailored for the multimodal corneal data. This layer effectively encodes the integrated features into a format that the transformer can process, accommodating the unique characteristics of each modality (e.g., spatial features from topography and temporal features from pachymetry). The images from corneal topography are preprocessed and encoded as sequences of patches, using the ViTs technique for handling image data. Time-series data, such as pachymetry and aberrometry measurements, are represented as sequences of feature vectors, each capturing the temporal evolution of key metrics. Scalar data from biomechanical assessments are incorporated as additional feature vectors, ensuring all data is aligned temporally and ready for integration. Here are the steps of the input embedding layer construction for the multimodal corneal data.

1. Let's assume that the multimodal corneal dataset consists of N samples, and each sample has M different modalities. Each modality i (where $i=1, 2, \dots, M$) has a different number of features d_i . The data can be represented as: $X_i=\{x_{i1}, x_{i2}, \dots, x_{idi}\}$ for each modality i .
2. Apply a linear transformation (Embedding Layer) to map the integrated features into a common embedding space of fixed dimension d . This transformation is performed using a learnable weight matrix $W_i \in \mathbb{R}^{d_i \times d}$ and a bias vector $b_i \in \mathbb{R}^d$ for each modality: $E_i = \widehat{X}_i W_i + b_i$, where $\widehat{X}_i$ is the normalized feature in modality i , and $E_i \in \mathbb{R}^{d_i \times d}$ is the embedded representation of modality i .
3. The embedded features from all modalities are concatenated to form a unified input embedding vector for each sample: $E = [E_1; E_2; \dots; E_M]$, where $E \in \mathbb{R}^{(d_1+d_2+\dots+d_M) \times d}$. Here, $(;)$ denotes the concatenation operation.

3.3.2 Positional encoding

Transformers lack an inherent mechanism to process the order or position of elements within a sequence since they are designed to work with sets rather than sequences. In the context of keratoconus progression detection, it is crucial to capture trends and changes over time (e.g., temporal sequences of measurements) or spatial relationships (e.g., different regions of corneal topography). Without positional encoding, the model would see the measurement data as a set of numbers without any notion of time order. For example, with positional encoding, the model recognizes that a curvature measurement at time t_1 precedes one at t_2 , allowing it to identify trends such as increasing curvature that may indicate keratoconus progression. Hence, positional encoding is integrated into the transformer architecture to retain the sequential order of the data points. This technique encodes the position of each element in the input sequence, allowing the transformer to differentiate between features based on their temporal context. This is crucial for enabling it to capture complex temporal and spatial dependencies that are essential for accurate keratoconus detection and progression analysis. Given an input sequence length L and a model embedding dimension d , positional encoding generates a matrix $PE \in \mathbb{R}^{L \times d}$ where each position k in the sequence has a unique encoding. A common approach is the sinusoidal positional encoding, defined as follows:

For each position k (where $k=0, 1, 2, \dots, L-1$) and each dimension i (where $i=0, 1, 2, \dots, d-1$):

$$PE(k, 2_i) = \sin\left(\frac{k}{10000^{\frac{2_i}{d}}}\right)$$

$$PE(k, 2_i + 1) = \cos\left(\frac{k}{10000^{\frac{2_i}{d}}}\right)$$

Where, $PE(k, 2_i)$ and $PE(k, 2_i + 1)$ represent the positional encoding values for even and odd dimensions, respectively. The use of sine and cosine functions of different frequencies ensures that each position k has a unique encoding. These functions allow the model to capture both local and global position information by varying across different dimensions. The denominators $10000^{2_i/d}$ control the frequencies of the sine and cosine waves, ensuring that the positional encodings vary smoothly across positions. After computing the positional encodings, they are added to the input embedded features to incorporate position information into the model. Let $E \in \mathbb{R}^{L \times d}$ be the original input embedding: $E' = E + PE$. Where, E' is the final input to the transformer. Each position k in the input sequence now contains information about both the feature values and their respective positions within the sequence. Hence, the final input embedding E' encapsulates the multimodal data in a unified representation that captures the complexities and nuances of each modality.

3.3.3 transformer encoder layers

The core of the classifier consists of multiple stacked transformer encoder layers. The encoder layers are responsible for extracting features and capturing dependencies within the input data. Each encoder layer contains two sub-layers: a multi-head self-attention mechanism and a position-wise fully connected feed-forward network.

3.3.3.1 Multi-head attention The multi-head attention mechanism is central to the model's ability to integrate and analyze diverse corneal features effectively, leading to a more robust and interpretable prediction of keratoconus stages. By having multiple heads, the model can attend to different parts of the data simultaneously, focusing on various aspects of corneal health, such as shape, thickness, and biomechanical properties. The mechanism identifies complex and nonlinear dependencies across integrated multimodal data, capturing interactions that are essential for accurate keratoconus detection and classification. It produces a more comprehensive representation of the corneal condition by allowing cross-talk between different data modalities, thereby facilitating the model's ability to make more nuanced classifications. Here is a formulaic description of the multi-head attention mechanism and its role in capturing complex dependencies across multimodal corneal data.

1. Input and linear transformation: Given the integrated input embedding $E \in \mathbb{R}^{L \times d}$, where L is the sequence length (the number of features after concatenation) and d is the embedding dimension. The multi-head attention mechanism first transforms E into

- three different matrices: queries Q , keys K , and values V . These matrices are obtained by linear transformations using learnable weight matrices $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$: $Q = EW_Q$, $K = EW_K$, $V = EW_V$. Where, d_k is the dimensionality of the query and key vectors.
2. Scaled dot-product attention: For each head h , the attention mechanism calculates the attention scores between the query and key vectors to determine the importance of each feature:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Where, QK^T computes the dot product between the query and key vectors, which gives the raw attention scores indicating how much focus one feature should have on another. $\frac{1}{\sqrt{d_k}}$ scales these scores to avoid exceedingly large values that can destabilize gradients during training. The softmax function normalizes the scores to a probability distribution, highlighting which features are most relevant. The attention scores are then used to weight the value vectors V , resulting in a new set of weighted features that capture interactions across the entire input.

3. Multi-head attention: The multi-head attention mechanism uses multiple attention heads to capture different aspects of the feature interactions. If there are h heads, each head i computes its own attention: $\text{head}_i = \text{Attention}(Q_i, K_i, V_i)$, where Q_i, K_i, V_i are projections of the input embedding using different weight matrices: $Q_i = EW_Q$, $K_i = EW_K$, $V_i = EW_V$. These heads are then concatenated and linearly transformed to form the final output: $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) WO$, where $WO \in \mathbb{R}^{hd_k \times d}$ is the weight matrix for the final linear transformation.
4. Capturing dependencies across modalities: The multi-head attention mechanism's primary role in this context is to capture complex dependencies and interactions across the different modalities of corneal data:
 - Cross-modality relationships: By attending to various parts of the integrated feature vector, each head can focus on different interactions. For example, one head might capture the relationship between corneal curvature (from topography) and corneal thickness (from pachymetry), while another head might focus on the interaction between biomechanical properties and aberrometry data.
 - Temporal and spatial patterns: In the case of time-series or spatial data (e.g., temporal sequences of corneal measurements), multi-head attention can learn how changes in one modality over time relate to changes in another, thereby identifying patterns crucial for early detection of keratoconus progression.
 - Enhanced feature representation: By concatenating the outputs of multiple attention heads, the model gains a richer representation of the data, incorporating a broader set of features and their interdependencies. This allows the model to capture subtle nuances and complex patterns that single-head attention mechanisms or traditional models might miss.

5. Output of multi-head attention: The output of the multi-head attention mechanism is a new set of features where each original feature vector is now represented in terms of its relationships with all other features across modalities. This output is then processed by subsequent layers (e.g., feed-forward layers, normalization) in the transformer to further enhance its predictive capacity.

3.3.3.2 Feed-forward neural network The Feed-Forward Neural Network (FFNN) layer in the transformer architecture plays a critical role in the model by enhancing the representational capacity of the transformer, enabling it to make sense of complex and non-linear relationships in the multimodal corneal data. While the attention mechanism captures relationships and dependencies between features, the FFNN layer further processes these relationships to extract complex patterns and make more informed predictions. Each position in the sequence (each feature) is independently processed through the FFNN layer, allowing the model to learn more abstract features from the integrated corneal data, ultimately improving the accuracy of keratoconus detection. The FFNN layer in the transformer is applied to each position separately and identically. It consists of two linear transformations with a non-linear activation function in between. Given an input $x \in \mathbb{R}^d$ (where d is the embedding dimension): $FFNN(x) = \max(0, xW_1 + b_1)W_2 + b_2$.

Where, W_1 and $W_2 \in \mathbb{R}^{d \times d_{ff}}$ are the weight matrices, $b_1 \in \mathbb{R}^{d_{ff}}$ and $b_2 \in \mathbb{R}^d$ are the bias vectors, d_{ff} is the dimensionality of the hidden layer in the FFNN, typically larger than d (e.g., $d_{ff} = 4d$) to allow the network to learn more complex patterns.

3.3.3.3 Layer normalization and dropout Layer normalization is applied after each sub-layer (e.g., multi-head attention and feed-forward network) in the transformer. This means that the output of each sub-layer is normalized before being passed to the next layer. This helps maintain stable activations and gradients throughout the network. In addition, layer normalization is used to stabilize and accelerate the training process by normalizing the inputs to each layer, preventing issues like vanishing or exploding gradients, especially in deep networks like transformers. Normalization ensures that the activations of each layer are on a consistent scale, which can lead to improved generalization, reducing the risk of overfitting. Since layer normalization operates independently for each input data point, it is particularly well-suited for transformers, which process sequences rather than batches of data. Given an input vector $x \in \mathbb{R}^d$, where d is the dimensionality of the features, layer normalization is computed as follows: First, compute the mean μ and variance σ^2 of the input across the feature dimension, then, normalize the input using the computed mean and variance:

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + c}}$$

where c is a small constant added for numerical stability. Finally, apply learnable scaling and shifting parameters γ and β to allow the network to adjust the normalized output,

$y_i = \gamma \hat{x}_i + \beta$, where γ and β are learned parameters, and y_i is the final output of the layer normalization for the i -th feature.

On other hand, dropout is a regularization technique used to prevent overfitting by randomly “dropping out” (setting to zero) a subset of neurons during training. This forces the network to learn more robust and generalized features. Given an input vector x and a dropout rate p (the probability of setting a neuron to zero), during training, generate a random binary mask $m \in \{0, 1\}^d$, where each element is drawn from a Bernoulli distribution with probability p : $m_i \sim \text{Bernoulli}(1 - p)$. Element-wise multiply the input x with the mask m to obtain the dropped-out output: $\hat{x}_i = m_i \cdot x_i$. To keep the output’s expected value consistent during both training and inference, scale the output during training:

$$\tilde{x} = \frac{1}{1 - p} (m \odot x)$$

where $\odot$ denotes element-wise multiplication. By randomly dropping out neurons during training, dropout prevents the network from relying too heavily on specific features, forcing it to learn more generalizable patterns. Dropout encourages the network to learn redundant representations of the data, ensuring that important features are captured by multiple neurons rather than a single pathway. During inference, all neurons are used, but they have learned to function under conditions where some inputs are missing, leading to a more robust model.

4 Experimental evaluation

Python version 3.12.4 was used to implement the suggested model. In this section, we will compare the results of the proposed methodology with those of three other recent approaches: SVM-based keratoconus detection (Shanthi et al. 2020), random forests-based keratoconus detection (Cohen et al. 2022), and CNN-based keratoconus detection (Kallel et al. 2024), which was previously discussed in the [literature review](#) section.

4.1 Dataset

Finding a comprehensive keratoconus dataset that includes all the data modalities, such as OCT images, corneal topography images, aberrometry, pachymetry, and other biomechanical data, is challenging due to the specialized nature of this medical data. Therefore, the proposed approach was evaluated using two different datasets of real-world healthcare data: The Keratoconus detection dataset (Keratoconus detection dataset 2024) and the Tabular Dataset for Keratoconus (Tabular Dataset for Keratoconus 2024). The first set includes a set of corneal topography images, and the second set includes a set of biomechanical data such as aberrometry and corneal thickness obtained from OCT-based topography electronic devices. The corneal biomechanical properties refer to the structural and mechanical characteristics of the cornea, the transparent front part of the eye (Marinescu et al. 2022). These properties include the cornea’s elasticity, stiffness, and ability to resist deformation under various forces. They are crucial for maintaining the cornea’s shape and clarity, which are essential for proper vision. Abnormalities in these properties can lead to conditions such as

keratoconus, where the cornea becomes too thin and weak, causing it to bulge outward and distort vision. Understanding and measuring corneal biomechanics is vital for diagnosing and managing such disorders, as well as for planning and evaluating surgical interventions like refractive surgery.

Keratoconus Detection Data Set This collection of corneal images is collected using Scheimpflug imaging systems (Pentacam). The Pentacam is a sophisticated diagnostic device used in ophthalmology to create detailed three-dimensional images of the anterior segment of the eye, including the cornea, iris, and lens (Mohammadpour and Heidari 2021). It employs a rotating Scheimpflug camera that captures multiple cross-sectional images of the eye, which are then compiled to generate precise topographic maps. These maps provide critical information about corneal thickness, curvature, and elevation, as well as the anterior and posterior corneal surfaces. The Pentacam is particularly valuable in diagnosing and managing conditions like keratoconus, evaluating candidates for refractive surgery, and monitoring corneal changes over time. Its non-invasive nature and high accuracy make it a standard tool in eye care.

On the other hand, each of the collected images has seven features. The seven features are: the corneal thickness (pachymetry map), the anterior eccentricity of the cornea, the posterior eccentricity of the cornea, the anterior cornea curves outward, the posterior cornea curves outward, the anterior curvature of the cornea, and the posterior curvature of the cornea. The entire number of records is 573 records. The dataset was divided into an approximate 74%:26% ratio of the training and testing data sets, respectively. In this case, the training dataset has 423 records, of which 150 are keratoconus infection records, 150 are normal eye records, and 123 are suspected to have keratoconus infection records, of which each corneal has 7 maps. While the test dataset has 150 records, 50 of which are infection records, 50 are normal eye records, and 50 are suspected to have infection records, each corneal also has 7 maps. Moreover, to achieve the authenticity of the evaluations, there are no redundant records among the training and testing sets.

Tabular Dataset for Keratoconus: The Keratoconus stage dataset is a new generation of datasets for evaluating the accuracy and effectiveness of various deep learning-based Keratoconus progression detection systems. The datasets include 446 features of corneal topography to distinguish between normal eye and corneal disease. The entire number of records is 3080, of which 264 are keratoconus infection records, 2595 are normal eye records, and 221 are suspected to have keratoconus infection records. The dataset was divided into an approximate 70%:30% ratio of the training and testing data sets, respectively. In this case, the training dataset has 2156 records, while the test dataset has 924 records. Also, to achieve the authenticity of the evaluations, there are no redundant records among the training and testing sets.

It is worth noting that the diagnosis of keratoconus cannot rely solely on a single curvature metric due to the complex and progressive nature of the disease. As noted, keratoconus diagnosis benefits from comprehensive multimodal data integration, and in our study, we have utilized various sources to achieve a robust classification system, including corneal topography, aberrometry, pachymetry, and biomechanical properties, to address this complexity. The distinction between definite keratoconus, suspected keratoconus, and normal corneas is drawn based on a combination of these multimodal metrics, processed through a transformer-based model designed to capture both spatial and temporal relationships within the data. Specifically:

1. Corneal topography and pachymetry: These provide detailed morphological insights, such as anterior and posterior corneal curvature and corneal thickness distribution. Changes in these metrics are critical in identifying different stages of keratoconus (Bui et al. 2023). For example, corneal thinning, combined with increased curvature, is more indicative of definite keratoconus, while milder deviations from normal in these parameters may point to suspected keratoconus.
2. Aberrometry and biomechanical properties: These offer additional diagnostic dimensions. Aberrometry provides information about optical imperfections caused by corneal irregularities, and biomechanical properties (e.g., corneal stiffness) help assess the structural integrity of the cornea (Marinescu et al. 2022). These measurements are particularly useful in detecting early or subclinical stages of keratoconus that are not as evident from topography alone.

Therefore, the use of a transformer-based model allows for the integration of these diverse data sources, ensuring that the relationships between them are taken into account in the classification process. The self-attention mechanism of transformers helps the model focus on the most informative features, considering interactions across different data modalities. Hence, this enables the model to distinguish between normal corneas, suspected keratoconus, and definite keratoconus with a high degree of accuracy. Through this comprehensive approach, we ensure that the boundary between these classifications is not based on a single value but rather on the synergistic interpretation of multiple corneal features. The combination of these modalities and the sophisticated modeling techniques used make the classification more reliable and reflect the true complexity of keratoconus progression.

4.2 Assessment metrics

Evaluating the proposed model's performance is critical to validating its effectiveness. Selecting the appropriate validation metric is one of the most essential aspects that must be considered. Therefore, the accuracy of a classifier, precision, recall, and F-score were utilized as standard assessment metrics in our experiments (Hossin and Sulaiman 2015). The classifier's accuracy is calculated as the total number of cases divided by the number of instances that were successfully identified. Precision, or positive predictive value, is the likelihood that the model will properly classify any entity. The recall rate is the percentage of test outcomes that the model properly detects; it is most likely referred to as sensitivity. The F-score is the harmonic mean of precision and recall. Moreover, the performance of the proposed transformer-based model will be compared to the existing state-of-the-art methods for keratoconus detection to demonstrate its superiority.

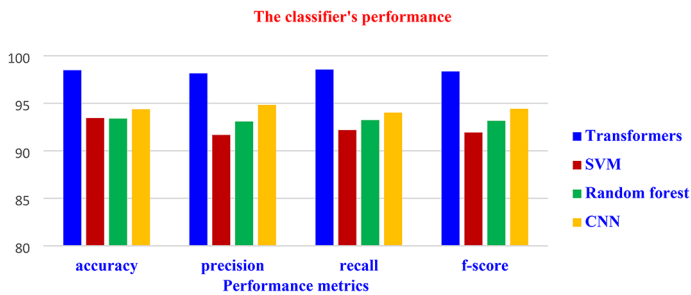
$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

Table 1 The classifier's performance metrics based on the "Keratoconus Detection" dataset

	Accuracy	Precision	Recall	F-score
Transformers	98.48	98.15	98.55	98.35
SVM	93.45	91.67	92.18	91.92
Random forest	93.39	93.08	93.23	93.15
CNN	94.37	94.83	94.02	94.42

**Fig. 4** A comparison of performance evaluation measures of the proposed methodology and three other recent approaches based on the "Keratoconus Detection" dataset

$$F\ score = \frac{2 * (Recall * precision)}{(Precision + recall)}$$

4.3 The experiments

A set of experiments on both the "Keratoconus Detection" and "Tabular Dataset for Keratoconus" datasets were conducted to compare the performance evaluation measures of the proposed methodology and three other recent approaches: "SVM-based," "random forests-based," and "CNN-based" keratoconus detection. Table 1; Fig. 4 demonstrate the classification results for the "Keratoconus Detection" dataset.

Moreover, to check the effect of varying the size of the data set on the experimental outcomes, the experiments were repeated with different sample sizes for each performance measure, as shown in Tables 2, 3 and 4, and 5 for the "Tabular Dataset for Keratoconus."

As shown in Tables 1, 2, 3, 4 and 5, the proposed methodology outperforms other related methodologies in terms of accuracy, which shows how our system is more powerful in detecting Keratoconus progression. Also, the proposed methodology achieved higher precision, which reflects the ability to detect the correct classification. Also, the proposed methodology achieved a higher recall score than other methodologies, which reflects the ability to classify the Keratoconus instances better than others. Finally, the results show that the proposed methodology has a higher value of F-score, which indicates that it is performing better on recall and precision.

These experimental results require in-depth analysis and discussion to demonstrate the reasons why the transformer-based approach outperforms other related approaches like SVM-based, Random Forests-based, and CNN-based models. We need to consider several aspects of machine learning models: their architecture, the nature of the data they process, and the specific task of keratoconus detection. Here is a detailed comparison and analysis of some key factors:

Table 2 The classifier's accuracy with different dataset sizes and based on the "Tabular dataset for Keratoconus"

Data size	500	1000	1500	2000	2500	3080	Average
Transformers	92.48	93.15	94.55	97.85	98.83	99.19	96.008
SVM	89.45	91.67	92.18	92.76	93.03	93.65	92.123
Random forest	90.39	92.08	93.23	93.59	93.99	94.39	92.945
CNN	91.37	92.83	93.52	94.45	95.26	95.84	93.878

Table 3 The classifier's precision with different dataset sizes and based on the "Tabular dataset for Keratoconus"

Data size	500	1000	1500	2000	2500	3080	Average
Transformers	91.83	93.36	94.84	97.95	98.83	99.21	96.003
SVM	88.93	90.46	91.72	93.08	93.79	94.33	92.052
Random forest	87.65	88.59	89.44	90.11	91.62	92.04	89.908
CNN	89.36	90.28	91.78	92.75	93.56	94.83	92.093

Table 4 The classifier's recall with different dataset sizes and based on the "Tabular dataset for Keratoconus"

Data size	500	1000	1500	2000	2500	3080	Average
Transformers	92.48	94.55	96.51	97.98	98.28	98.97	96.462
SVM	89.03	91.02	92.65	93.69	94.09	94.16	92.440
Random forest	90.12	91.53	92.85	92.87	93.11	93.17	92.275
CNN	89.83	90.26	91.98	93.44	94.7	94.94	92.525

Table 5 The classifier's f-score with different dataset sizes and based on the "Tabular dataset for Keratoconus"

Data size	500	1000	1500	2000	2500	3080	Average
Transformers	92.15385	93.95123	95.66771	97.965	98.55423	99.08985	96.230
SVM	88.97997	90.73914	92.18265	93.384	93.93976	94.24492	92.245
Random forest	88.86784	90.03601	91.11311	91.46918	92.35899	92.60155	91.074
CNN	89.59438	90.27	91.87989	93.09372	94.12655	94.88497	92.308

1. Transformers excel in integrating and processing multimodal data due to their flexible architecture, which can accommodate diverse data types and capture complex relationships across them. This is crucial for keratoconus detection, where data from various sources must be integrated to form a comprehensive understanding. Compared to other traditional models like SVMs or Random Forests, which are not inherently designed to handle multimodal data, it is challenging to integrate diverse data sources like those used in keratoconus detection. Hence, they are generally less effective at capturing complex, non-linear relationships in data. Also, while CNNs excel at processing images, they are less suited for integrating and processing other types of data without significant architectural modifications.
2. Transformers are adept at handling sequential data, making them well-suited for time-series analysis and scenarios where the temporal relationship between data points is important. In the context of keratoconus detection, this can help in tracking the progression of the disease over time. Compared with the traditional models like Random forests or CNN, which are not naturally suited for sequential data, limiting their effectiveness

- in tasks requiring temporal analysis, such as monitoring keratoconus progression over time.
3. The self-attention mechanism allows transformers to capture both temporal and spatial dependencies within the data. This capability is particularly beneficial for tracking keratoconus progression, as it involves understanding changes over time and space.
 4. Transformers can learn hierarchical feature representations, enabling them to capture subtle and complex patterns in keratoconus data that other models might miss. This is especially important for detecting early-stage keratoconus, where changes are often subtle.
 5. Despite their computational demands, transformers are highly parallelizable and can scale efficiently with data size, allowing them to train on larger datasets, achieve faster model convergence, and generalize better than traditional models like SVMs or Random Forests. Which struggle with large-scale datasets due to their computational complexity and memory usage.
 6. The architecture of transformers is inherently more adaptable to complex, high-dimensional problems compared to traditional models. This adaptability allows them to model the intricate interactions between various features present in Keratoconus datasets.

In conclusion, the transformer-based approach offers significant advantages over SVM-based, Random Forests-based, and CNN-based models for keratoconus detection due to its ability to handle multimodal data, handle sequential data, capture temporal and spatial dependencies, capture complex patterns, and provide scalable and efficient learning. While each traditional model has its strengths, the unique capabilities of transformers make them particularly well-suited for the task of keratoconus detection, where diverse data sources and subtle patterns must be integrated and analyzed effectively. By leveraging these strengths, the transformer-based approach not only enhances the accuracy and reliability of keratoconus detection but also sets the stage for future advancements in medical Artificial Intelligence (AI) applications, where the integration of diverse data sources and the modeling of complex relationships are increasingly critical.

4.3.1 The experimental results visualization

As shown in Figs. 5, 6, 7 and 8, a comparison is made between the performances of the proposed methodology and “SVM-based,” “Random forests-based,” and “CNN-based” keratoconus detection through a different set of experiments with different measures, in which the x-axis represents the data set size and the y-axis denotes the performance measure. Some experiments were applied to different sample sizes to verify the effect of varying data set sizes on the experimental results. As we note, with the small data volume, the model was not trained enough to be able to show efficient performance (i.e., 500–1500). This indicates that the model was not trained enough to meet our expectations (low training errors and large testing errors). When the training data volume reaches a sufficient volume to allow the model to improve its performance, we will observe an improvement in the model’s performance as the data volume increases (i.e., 2000–3080). Furthermore, it can be observed that the proposed approach outperforms other related work approaches across all measures.

As the experiment results demonstrate, integrating data from multiple sources, such as corneal tomography, pachymetry, and biomechanical properties, has shown promise in

Fig. 5 A comparison of the accuracy for Transformers-based, SVM-based, Random forests-based, and CNN-based keratoconus detection with different sample sizes

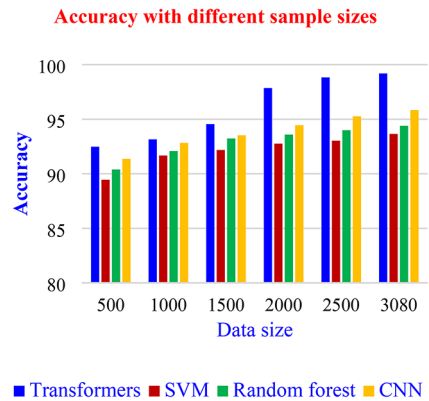


Fig. 6 A comparison of the precision for Transformers-based, SVM-based, Random forests-based, and CNN-based keratoconus detection with different sample sizes

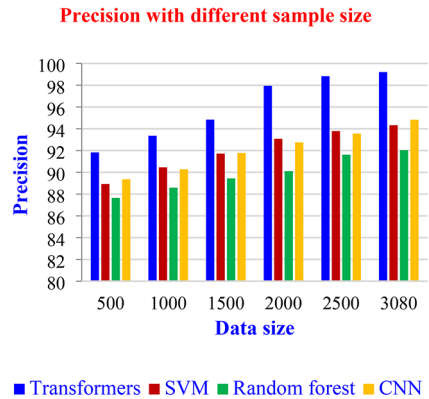
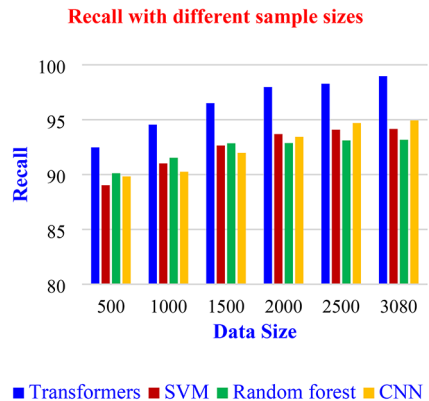
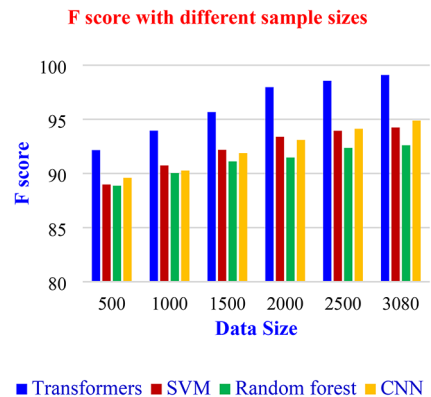


Fig. 7 A comparison of the recall for Transformers-based, SVM-based, Random forests-based, and CNN-based keratoconus detection with different sample sizes



enhancing detection accuracy. The proposed approach outperformed traditional methods by capturing complementary information from different modalities, leading to more accurate and robust detections. Thus, the effective use of different data sources and data types will help improve disease detection accuracy. In addition, the proposed approach may help phy-

Fig. 8 A comparison of the recall for Transformers-based, SVM-based, Random forests-based, and CNN-based keratoconus detection with different sample sizes



sicians assess corneal conditions and detect keratoconus, which is otherwise difficult to do using subjective judgements, especially in the preclinical and early stages of the disease.

4.3.2 Execution time analysis

The execution time analysis is a critical component of evaluating the efficiency and scalability of different machine learning models. Therefore, Fig. 9 shows a comparison between the execution times of “Transformers-based,” “SVM-based,” “Random forests-based,” and “CNN-based” keratoconus detection based on the “Keratoconus Detection” data set. Where the x-axis represents the utilized methodology and the y-axis denotes the execution time in seconds. The results show that the proposed approach achieves the lowest execution time, significantly outperforming the other methods. This can be attributed to two factors: The set of techniques that were used in the data preprocessing phase, such as PCA, which achieved up to 62.5% dimensionality reduction; and the inherent parallelization capabilities of transformers, which allow them to process data more efficiently compared to the other models.

Furthermore, Fig. 10 shows the same comparison through a different set of experiments with different dataset sizes and based on “Tabular Dataset for Keratoconus” datasets. Where the x-axis represents the sample size and the y-axis represents the execution time in seconds. The results highlight the efficiency and scalability of the transformer-based model. As the dataset size increases, the execution time grows linearly, showcasing the model’s ability to handle larger datasets without a substantial increase in computational time. This scalability is crucial for real-world applications, where data volumes are continually expanding.

The SVM and Random Forests models show a more significant increase in execution time with larger datasets. The complexity of these models often leads to higher computational demands, particularly when dealing with high-dimensional data, as is the case with keratoconus detection. The lack of inherent parallelization in these methods further exacerbates the execution time as data size increases. In addition, although CNNs are known for their efficiency in processing image data, their performance here is slightly slower than transformers. This can be attributed to the overhead of convolutional operations, which, while effective for feature extraction, are computationally intensive.

The architecture of transformers, with their self-attention mechanisms and feedforward layers, allows them to process data in parallel, reducing execution time. This advantage is particularly noticeable when handling large datasets and multimodal data, as transformers

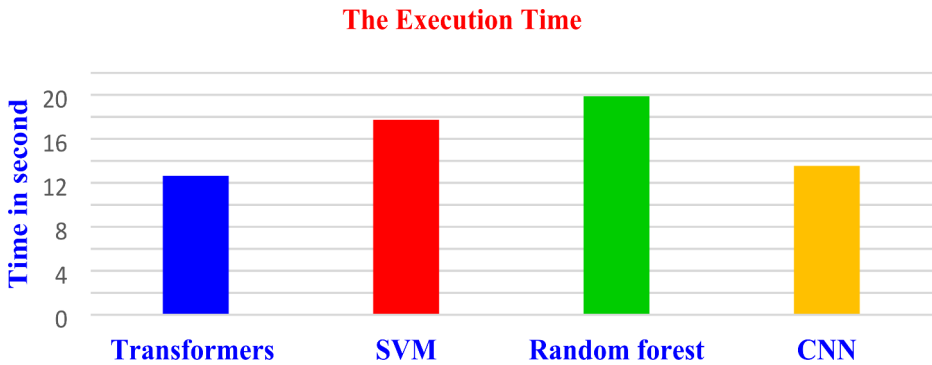


Fig. 9 A comparison of the execution times for Transformers-based, SVM-based, Random forests-based, and CNN-based keratoconus detection

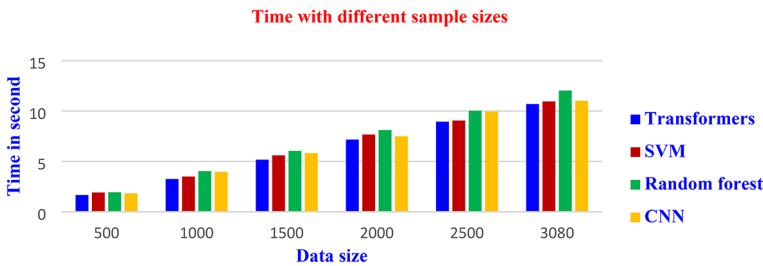


Fig. 10 A comparison of the execution times for Transformers-based, SVM-based, Random forests-based, and CNN-based keratoconus detection, with different sample sizes

can efficiently integrate and analyze diverse information sources. The lower execution time of transformers not only implies faster processing but also translates to reduced computational costs and energy consumption. In clinical settings, where timely decision-making is crucial, the ability to quickly analyze and interpret data is a significant advantage. In conclusion, the transformer-based model demonstrates superior execution time performance compared to SVM, Random Forest, and CNN models, making it a highly efficient choice for keratoconus detection. Its ability to handle large datasets efficiently, coupled with its superior accuracy and adaptability, positions it as a leading approach in the field of medical AI applications.

5 The computational complexity of the transformers model

Analyzing the computational complexity and resource requirements of the proposed transformer-based approach for keratoconus detection is crucial to understanding its practical implementation, scalability, and efficiency. The key components of transformers that contribute to their computational complexity are:

- **Self-Attention mechanism:** the self-attention mechanism allows transformers to weigh the importance of different elements within a sequence. For a given input sequence, this

mechanism computes a set of attention scores, which requires significant computational resources.

- Feedforward neural networks: each layer of a transformer includes a feedforward neural network that processes the attended features, adding to the computational load.
- Layer stacking: transformers consist of multiple layers stacked on top of each other, which increases their capacity to learn complex patterns but also contributes to higher computational complexity.

Here's a detailed discussion of the computational demands and resource considerations involved in using transformers for this task:

5.1 Computational complexity analysis

The computational complexity of transformers primarily arises from the self-attention mechanism, but the other layers also contribute. Let's break down this complexity:

- Self-Attention complexity: the self-attention mechanism has a complexity of $O(n^2 \cdot d)$, where n is the sequence length and d is the dimensionality of the model. This is due to the pairwise interactions between all elements in the sequence, which require computing dot products for each pair.
- Feedforward layer complexity: the complexity of the feedforward layers in each transformer block is $O(n \cdot d^2)$, as each element of the sequence is processed independently through a feedforward network.
- Overall complexity: for a transformer with L layers, the total complexity can be approximated as $O(L \cdot (n^2 \cdot d + n \cdot d^2))$. This reflects both the attention and feedforward operations for each layer.

5.2 Resource requirements

Given the computational complexity, transformers are resource-intensive models, requiring significant hardware resources for both training and inference:

- Memory requirements: transformers require substantial memory to store the activations and gradients during training, especially for long sequences. This can lead to high graphics processing unit or tensor processing unit memory usage.
- Processing power: training transformers demands powerful graphics processing units or tensor processing units to handle the large matrix multiplications involved in the attention mechanism and feedforward networks. The training time can vary significantly based on the hardware used and the size of the dataset.
- Data storage: handling multi-source data, such as images and tabular data, may require additional storage and preprocessing capabilities, which can increase the overall resource requirements.

5.3 Strategies to mitigate complexity

To manage the computational demands and resource requirements of transformer-based approaches, several strategies can be employed:

- Techniques such as Sparse Attention (Meister et al. 2021) and Linformer (Wang et al. 2020) have been proposed to reduce the complexity of the self-attention mechanism, bringing it down to $O(n \cdot d)$ or even lower by approximating attention scores.
- Reducing the dimensionality of input data through techniques like PCA or using lower-resolution images can decrease the computational load without significantly impacting model performance.
- Pre-trained transformer models can be fine-tuned on specific tasks, significantly reducing the training time and resources required compared to training from scratch.
- Utilizing techniques such as pruning and quantization (Singh et al. 2023) can reduce the size and complexity of the transformer networks model, making them more suitable for deployment on resource-constrained devices.

5.4 Comparison with the other models

- Compared to CNNs: CNNs are typically less resource-intensive than transformers for image-based tasks due to their localized operations and parameter sharing. However, they may not capture global dependencies as effectively as transformers.
- Compared to SVMs and Random Forests: traditional models like SVMs and Random Forests are generally more computationally efficient for small to medium-sized datasets but lack the capacity to handle complex, high-dimensional data as effectively as transformers.

5.5 Practical Considerations

- The scalability of transformers allows them to handle increasingly large datasets, but this requires access to advanced computing resources, such as distributed training across multiple graphics processing units or tensor processing units.
- There is often a trade-off between model performance and resource consumption. Optimizing transformer architectures and selecting appropriate hyperparameters can help balance these factors.
- Deploying transformer models for real-time applications may require optimizing inference speed, which can be achieved through model compression techniques and hardware accelerators.

In conclusion, the transformer-based approach for keratoconus detection offers significant advantages in terms of capturing complex, multimodal patterns, but it comes with substantial computational complexity and resource requirements. By leveraging efficient

transformer variants and optimizing model architecture, these challenges can be mitigated, enabling the practical application of transformers in clinical settings and real-time systems. Balancing performance and resource demands is essential to fully realizing the potential of transformers in medical AI applications.

6 Conclusion and future work

As demonstrated by significant advances in ophthalmic disease prediction over the past ten years, the use of ophthalmic digital pictures in conjunction with AI approaches holds enormous potential for enhancing the precision and efficiency of the identification, intervention, and treatment of keratoconus progression. AI can efficiently integrate medical data from various sources, including clinical examinations, symptoms, imaging data, patient-reported information, genetic information, and environmental factors, by utilizing deep learning models and advanced machine learning algorithms, which allows for early detection of vital patterns and symptoms related to keratoconus disease. Which assists healthcare practitioners in identifying individuals who face a heightened risk of developing the condition. In addition, by considering multiple factors, AI mechanisms can offer more holistic perceptions of the keratoconus progression, enabling individualized treatment approaches. This research presents a novel approach for the detection and prediction of keratoconus progression using transformer technology and multi-source data fusion. By integrating and analyzing diverse data sources, such as corneal topography, aberrometry, pachymetry, and biomechanical measurements, we have demonstrated the ability to accurately capture the complex patterns associated with keratoconus progression, thereby enabling timely interventions and improving patient outcomes. The transformer-based model outperforms traditional machine learning methods, such as SVM, Random Forests, and CNNs, in terms of accuracy, precision, recall, and F-score, while maintaining lower execution times, making it a highly efficient and effective tool for clinical settings.

The proposed multi-source detection system based on transformers represents a significant advancement in the field of keratoconus management and addresses the limitations of existing diagnostic techniques, which often rely on single-source data and subjective assessments, by offering a comprehensive and objective framework for early detection and management of keratoconus. This innovative approach not only enhances the accuracy and reliability of keratoconus progression detection but also holds significant potential for application in other ocular conditions and medical domains where multi-source data integration is crucial. Finally, it's worth mentioning that in this research, we have used a set of impressive state-of-the-art techniques at every stage of processing.

Building upon the promising results of this study, future research will explore several avenues to further enhance the performance and applicability of the proposed system:

1. Incorporation of more intuitive and objective indicators: introducing more intuitive and quantitative indicators for keratoconus classification with the aim of improving the model's predictive capabilities.
2. Real-time detection and monitoring: developing a real-time version of the proposed system that can be implemented in clinical practice, providing ophthalmologists with immediate insights into keratoconus progression and enabling timely interventions.

3. Collaborative research and data sharing: collaborative efforts among research institutions and clinical centers can facilitate the creation of large, diverse datasets. Standardizing data collection and annotation protocols can further improve model generalizability and reproducibility.
4. Transferability to other ocular diseases: extending the methodology to detect and monitor other ocular diseases, such as glaucoma and diabetic retinopathy, to assess the generalizability and adaptability of the transformer-based approach.
5. Robustness and generalizability: conducting further validation studies across diverse populations and clinical settings to evaluate the robustness and generalizability of the model, ensuring its applicability to a wide range of patients and healthcare environments.
6. Developing a clinical tool: creating a user-friendly interface for clinicians to use the model for early detection and management of keratoconus. Such as integrating the proposed model into OCT topography instruments.

By addressing these future research directions, the proposed system has the potential to significantly impact the field of ophthalmology, revolutionizing the early detection and management of keratoconus and other ocular diseases, thereby enhancing patient quality of life and advancing the state-of-the-art in medical machine learning.

Acknowledgements Not applicable.

Author contributions The manuscript is written by a single author, who contributed to all aspects of the manuscript.

Funding There are no sources of funding.

Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB).

Data availability The two data sets utilized during the current study are available in the Kaggle dataset repository, <https://www.kaggle.com/datasets>.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Alyaa H, Ghaeb N, Musa Z (2018) Support Vector Machine for Keratoconus detection by using Topographic maps with the help of Image Processing techniques. *IOSR Journal of Pharmacy and Biological Sciences* (IOSR-JPBS) 12.6, pp. 50–58

- Bui A, Truong A, Pasricha N, Indaram M (2023) Keratoconus diagnosis and treatment: recent advances and future directions. *Clin Ophthalmol* 17:2705–2718. <https://doi.org/10.2147/OPTH.S392665> PMID: 37736107; PMCID: PMC10511017
- Chen X, Zhao J, Iselin K et al (2021) Keratoconus detection of changes using deep learning of colour-coded maps. *BMJ Open Ophthalmol* 6:e000824. <https://doi.org/10.1136/bmjophth-2021-000824>
- Cohen E, Bank D, Sorkin N et al (2022) Use of machine learning to achieve keratoconus detection skills of a corneal expert. *Int Ophthalmol* 42:3837–3847. <https://doi.org/10.1007/s10792-022-02404-4>
- Feng R, Xu Z, Zheng X, Hu H, Jin X, Chen D, Yao K, Wu J (2021) KerNet: a Novel Deep Learning Approach for Keratoconus and Sub-clinical Keratoconus Detection based on Raw Data of the Pentacam HR System. *IEEE J Biomed Health Inf* 25(10):3898–3910. <https://doi.org/10.1109/JBHI.2021.3079430>
- Haddad R, Akansu A (1991) A class of fast Gaussian Binomial filters for Speech and Image Processing. *IEEE Trans Acoust Speech Signal Process* 39:pp723–727
- Han J, Pei J, Tong H (2022) *Data Mining: Concepts and Techniques*, 4th Edition, Morgan Kaufmann Publishers, Elsevier
- Hossin M, Sulaiman M (2015) A review on evaluation Metrics for Data classification evaluations. *Int J Data Min Knowl Manage Process* 5:01–11
- Jolliffe T, Cadima J (2016) Principal component analysis: a review and recent developments. *Philosophical Trans Royal Soc A: Math Phys Eng Sci* 374(2065):20150202. <https://doi.org/10.1098/rsta.2015.0202>. Bibcode:2016RSPTA.37450202J
- Kallel I, Mahfoudhi O, Kammoun S (2024) Deep learning models based on CNN architecture for early keratoconus detection using corneal topographic maps. *Multimed Tools Appl* 83:49173–49193. <https://doi.org/10.1007/s11042-023-17551-8>
- Kamiya K, Ayatsuka Y, Kato Y et al (2019) Keratoconus detection using deep learning of colour-coded maps with anterior segment optical coherence tomography: a diagnostic accuracy study. *BMJ Open* 9:e031313. <https://doi.org/10.1136/bmjopen-2019-031313>
- Kamiya K, Ayatsuka Y, Kato Y, Shoji N, Miyai T, Ishii H, Mori Y, Miyata K (2021) Prediction of keratoconus progression using deep learning of anterior segment optical coherence tomography maps. *Ann Transl Med* 9(16):1287. <https://doi.org/10.21037/atm-21-1772>
- Keratoconus detection dataset, accessed in July 20 (2024) <https://www.kaggle.com/datasets/elmechdi12/keratoconus-detection>
- Kuo B, Chang W, Liao T, Liu F, Liu H, Chu H, Chen W, Hu F, Yen J, Wang I (2020) Keratoconus screening based on deep learning approach of corneal topography. *Trans Vis Sci Tech* 9(2):53. <https://doi.org/10.1167/tvst.9.2.53>
- Lavric A, Valentin P (2019) KeratoDetect: Keratoconus Detection Algorithm using Convolutional neural networks. *Comput Intell Neurosci* 8162567:44–53. <https://doi.org/10.1155/2019/8162567>
- Lavric A, Popa V, Takahashi H, Yousefi S (2020) Detecting Keratoconus from Corneal Imaging Data Using Machine Learning, in *IEEE Access*, vol. 8, pp. 149113–149121, <https://doi.org/10.1109/ACCESS.2020.3016060>
- Linear interpolation *Encyclopedia of Mathematics*. URL: http://encyclopediaofmath.org/index.php?title=Linear_interpolation&oldid=27068
- Liu Y, Qiang W, Wen Q (2023) Artificial Intelligent Power forecasting for wind farm based on Multi-source Data Fusion. *Processes* 11(5):1429. <https://doi.org/10.3390/pr11051429>
- Marinescu M, Dascalescu D, Constantin M, Coviltir V, Burcel M, Darabus D, Ciuluvica R, Stanila D, Potop V, Alexandrescu C (2022) Corneal biomechanics - an emerging ocular property with a significant impact. *Maedica (Bucur)* 17(4):925–930. <https://doi.org/10.26574/maedica.2022.17.4.925>
- Meinard M (2007) Dynamic Time Warping. In *Information Retrieval for Music and Motion*, Chap. 4, pages 69–84 (PDF). Springer. <https://doi.org/10.1007/978-3-540-74048-3>
- Meister C, Lazov S, Augenstein I, Cotterell R (2021) Is Sparse Attention more Interpretable? Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Short Papers), pages 122–129
- Mohammadpour M, Heidari Z (2021) Pentacam. In: Mohammadpour M (ed) *Diagnostics in Ocular Imaging*. Springer, Cham. https://doi.org/10.1007/978-3-030-54863-6_3
- Niazi S, Jiménez-García M, Findl O, Gatziofias Z, Doroodgar F, Shahriari M, Javadi M (2023) Keratoconus diagnosis: from fundamentals to Artificial Intelligence: a systematic narrative. *Rev Diagnostics* 13:2715. <https://doi.org/10.3390/diagnostics13162715>
- Pudjihartono N, Fadason T, Kempa-Liehr A, O’Sullivan J (2022) A review of feature selection methods for machine learning-based Disease Risk Prediction. *Front Bioinform* 2:927312. <https://doi.org/10.3389/fbiof.2022.927312>

- Shanthi S, Nirmaladevi K, Pyingkodi M, Dharanesh K, Gowthaman T, Harsavardan B (2020) Machine learning approach for detection of keratoconus. IOP Conference Series: Materials Science and Engineering, Volume 1055, International Virtual Conference on Robotics, Automation, Intelligent Systems and Energy (IVC RAISE 2020), Erode, India
- Singh S, Sharma K, Karna B, Raj P (2023) Pruning and quantization for deeper Artificial Intelligence (AI) model optimization. In: Sharma S, Subudhi B, Sahu UK (eds) Intelligent Control, Robotics, and industrial automation. Lecture Notes in Electrical Engineering, vol 1066. Springer, Singapore. https://doi.org/10.1007/978-981-99-4634-1_73.
- Tabular Dataset for Keratoconus, accessed in August 1 (2024) <https://www.kaggle.com/datasets/snehakurmi/tabular-dataset-for-keratoconus>
- Thomas W, Lysandre D, Victor S, Julien C, Clement D, Anthony M, Pierric C, Tim R, Remi L, Morgan F, Joe D, Sam S, Patrick V, Clara M, Yacine J, Julien P, Canwen X, Teven L, Sylvain G, Mariama D, Quentin L, Alexander R (2020) Transformers: State-of-the-Art Natural Language Processing. Proceedings of the Conference on Empirical Methods in Natural Language Processing: System Demonstrations. pp. 38–45. <https://doi.org/10.18653/v1/2020.emnlp-demos.6>
- Usgaonkar U, Chodankar S, Shetty A (2023) Online survey about keratoconus management by optometrists. Indian J Ophthalmol 71(1):86–90. https://doi.org/10.4103/ijo.IJO_525_22
- Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez A, Kaiser Ł, Polosukhin I (2017) Attention is all you need. In Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS'17). Curran Associates Inc., Red Hook, NY, USA, 6000–6010
- Wan Q, Wei R, Ma K, Yin H, Deng YP, Tang J (2023) Deep learning-based automatic diagnosis of Keratoconus with corneal endothelium image. Ophthalmol Ther 12(6):3047–3065. <https://doi.org/10.1007/s40123-023-00795-w>
- Wang S, Li B, Khabsa M, Fang H, Ma H (2020) Linformer: Self-Attention with Linear Complexity, cite arxiv:2006.04768
- Wang M et al (2024) AI-Based Advanced Approaches and Dry Eye Disease Detection Based on Multi-Source Evidence: Cases, Applications, Issues, and Future Directions, in Big Data Mining and Analytics, vol. 7, no. 2, pp. 445–484, <https://doi.org/10.26599/BDMA.2023.9020024>
- Yousefi S, Yousefi E, Takahashi H, Hayashi T, Tampo H, Inoda S, Eguchi S, Tanaka T (2018) Keratoconus severity identification using unsupervised machine learning. PLoS ONE 13(11):e0205998.
- Zhang C, Lu J, Zhao Y (2024) Generative pre-trained transformers (GPT)-based automated data mining for building energy management: Advantages, limitations and the future, Energy and Built Environment, Volume 5, Issue 1, Pages 143–169, ISSN 2666–1233, <https://doi.org/10.1016/j.enbenv.2023.06.005>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.